

TOI-2046b

R-2041

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_251007 · created 2026-08-08T22:54:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460955.91 +/- 0.19 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.046
Transit depth (Rp/Rs)^2	1.23 +/- 1.03 [%]
Orbital Inclination (inc)	85.76 +/- 2.9
Airmass coefficient 1 (a1)	0.5183 +/- 0.0053
Airmass coefficient 2 (a2)	0.47 +/- 0.11
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	None
Optimal Aperture	2.77
Optimal Annulus	4.9
Transit Duration (day)	0.018 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.046 vs 0.1213 (deviation 0.22σ)
Duration fit vs published	0.018 d vs 0.1033 d (deviation 2.94σ)
Mid-time O–C	113.8 min
Depth SNR	2.4
Residual scatter	1.03 %

Light curve

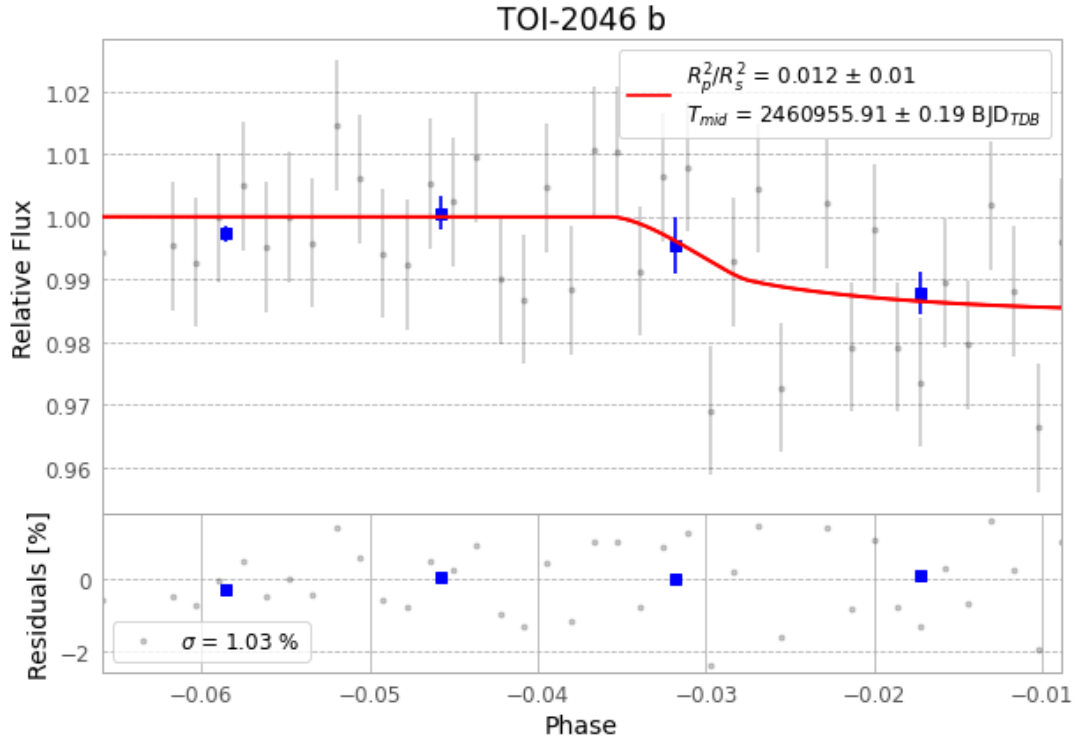


Figure 1 — FinalLightCurve_TOI-2046 b_2025-10-07.png (SHA-256 a07e6d9c1aebce8a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [380, 325], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 21d193c2998a09ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

40 FITS frames (26 MB), TOI2046251007072712.FITS → TOI2046251007093017.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-251007.log (824424911ea3...), FinalLightCurve_TOI-2046 b_2025-10-07.png (a07e6d9c1aeb...), FinalLightCurve_TOI-2046 b_2025-10-07.csv (a7984fe2e947...)

Compute resources

Wall time 121.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 22:54Z.